Connectome-Constrained Sensorimotor Behaviors and Modality-Specific Motor Channels in Drosophila

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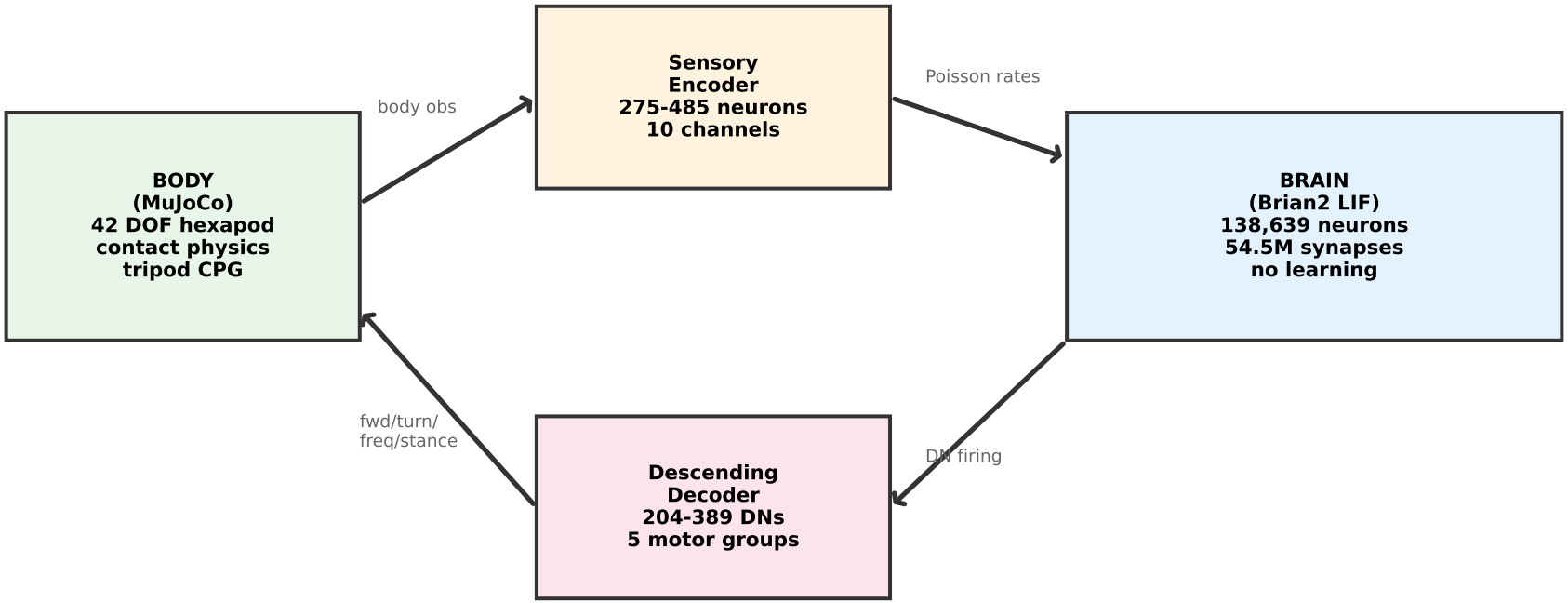
FlyWire connectome (138,639 neurons, 54.5M synapses)
Brian2 LIF simulation + MuJoCo biomechanical body
No learning, no parameter tuning

Abstract

The Drosophila melanogaster FlyWire connectome provides a neuron-level wiring diagram of 139,255 neurons and approximately 50 million synapses, but whether connectome-structured dynamics can support meaningful sensorimotor transformations without learning or task-specific optimization remains unknown. We built a closed-loop system coupling a Brian2 LIF simulation of 138,639 neurons (FlyWire 783 completeness snapshot; 15M connection pairs, 54.5M synapses) to a MuJoCo biomechanical fly body (FlyGym), with biologically identified sensory populations encoding stimuli and descending neuron populations decoding motor commands through an interpretable sensorimotor interface. The connectome-structured brain model, when coupled to an embodied fly through this transparent interface, produces adaptive and behaviorally specific responses without learning or parameter fitting: causal locomotion control (10/10 ablation tests, forward drive reduced 53% by targeted silencing), olfactory valence discrimination (opposite turning for attractive vs aversive odors, 6/6 tests), and visually guided looming escape (contralateral turning with escape index 1.11, abolished 21-fold in shuffled connectome controls). Analysis of the descending neuron populations underlying these behaviors reveals a structural principle: at the direct sensory-to-motor interface, modalities maintain near-complete segregation (Jaccard index 0.007-0.074), while convergence onto shared motor neurons occurs one synapse deeper (Jaccard 0.64-0.80) -- the connectome implements modality-specific labeled lines that merge through a single layer of interneurons.

Figure 1. Closed-loop connectome-body interface



Sensory channels: gustatory(20) | proprioceptive(33) | mechanosensory(15) | vestibular(7) | olfactory(100+100) | visual(50+50) | LPLC2(108+102)

Decoder groups: forward(59) | turn_left(136) | turn_right(167) | rhythm(25) | stance(25)

Figure 2. Causal locomotion control from connectome wiring

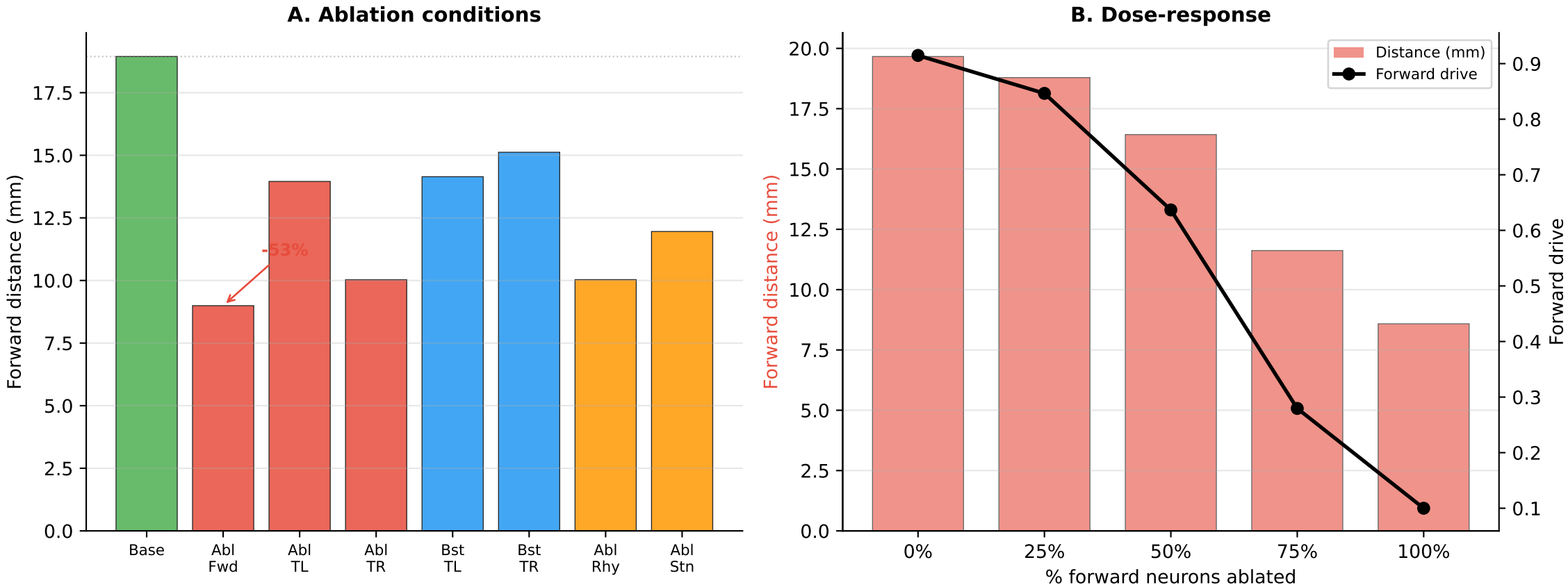
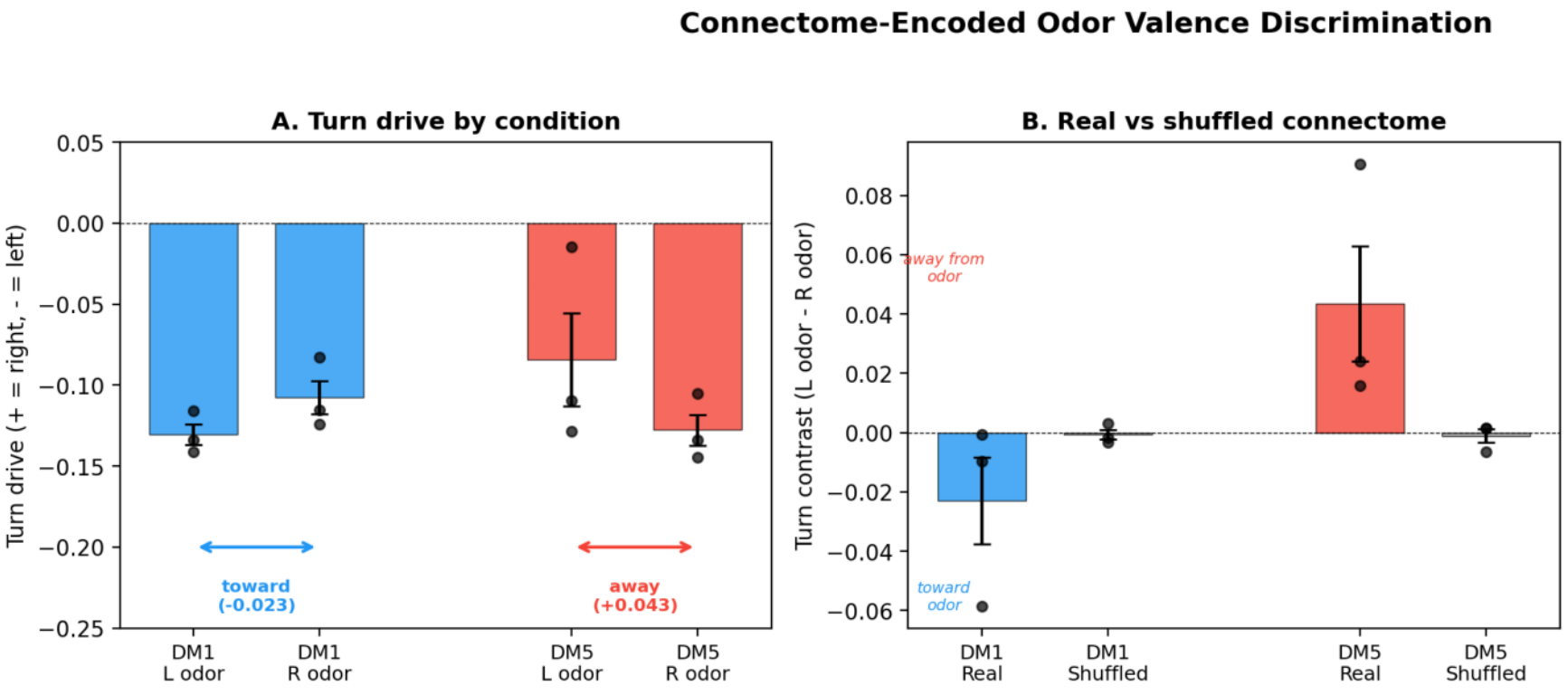


Figure 3. Connectome-encoded odor valence discrimination

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Or42b vs Or85a Odor Valence
139,000-neuron connectome (FlyWire)
204 descending readout neurons
3 seeds per condition

REAL CONNECTOME
  DM1 (Or42b) contrast: -0.0229
  DM5 (Or85a) contrast: +0.0435
  Valence contrast:     +0.0664

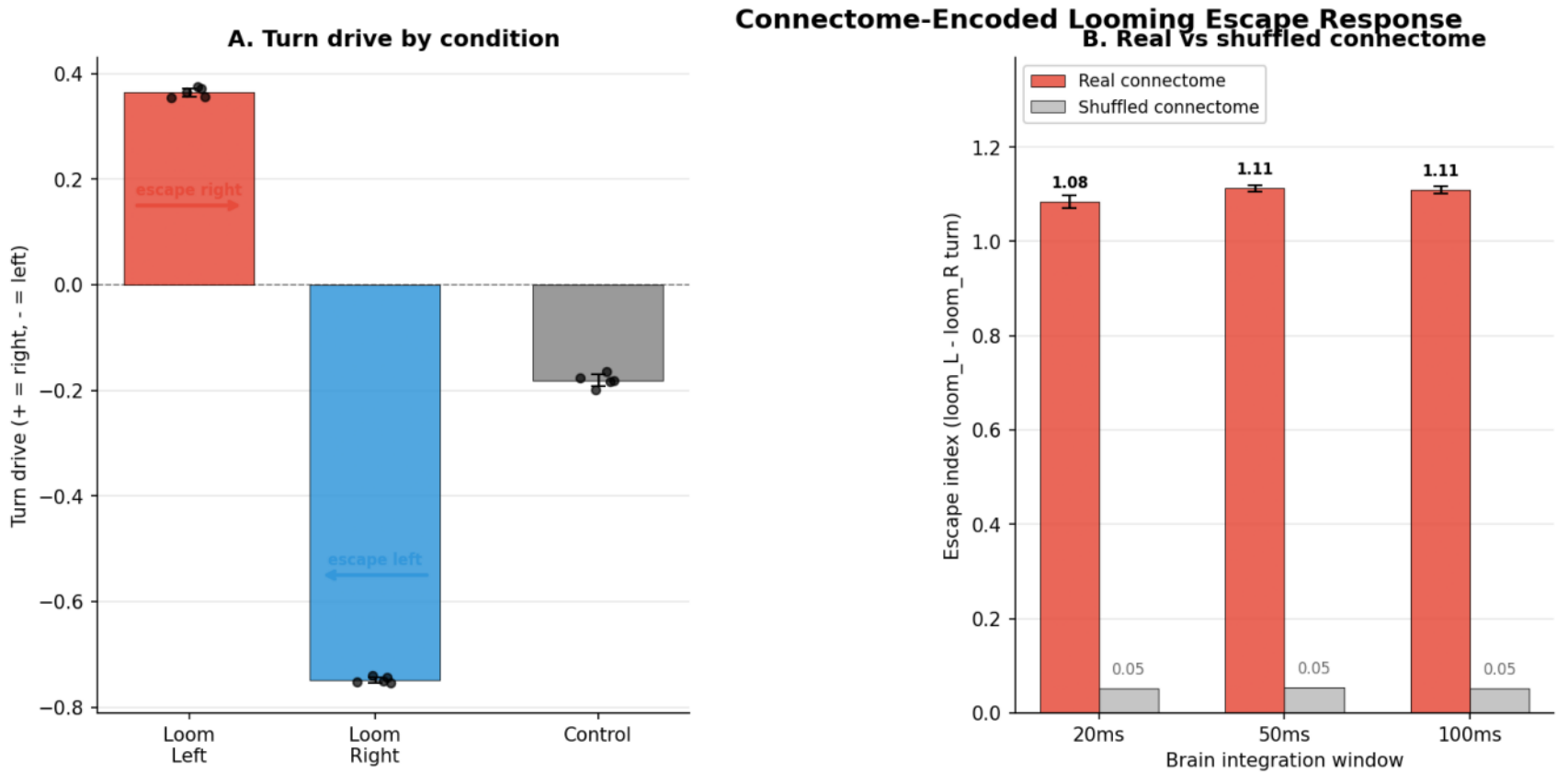
SHUFFLED CONNECTOME
  DM1 contrast: -0.0005
  DM5 contrast: -0.0011
  Valence contrast: -0.0005
```

[PASS] DM1_attractive_turning
[PASS] DM5_aversive_turning
[PASS] opposite_valence
[PASS] connectome_specificity
[PASS] DM1_connectome_specificity
[PASS] DM5_connectome_specificity

6/6 tests passed

Drosophila FlyWire connectome (139k neurons, 15M synapses) | Brian2 LIF simulation | No learning

Figure 4. Connectome-encoded looming escape response

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LPLC2 Looming Escape Circuit
139,000-neuron connectome (FlyWire)
LPLC2: 108L + 102R neurons
389 descending readout neurons
5 seeds per condition

REAL CONNECTOME (50ms)
  Loom L turn:  +0.364
  Loom R turn:  -0.748
  Control turn: -0.181
  Escape index: 1.112

SHUFFLED CONNECTOME (50ms)
  Escape index: 0.0530
  (21x weaker)
```
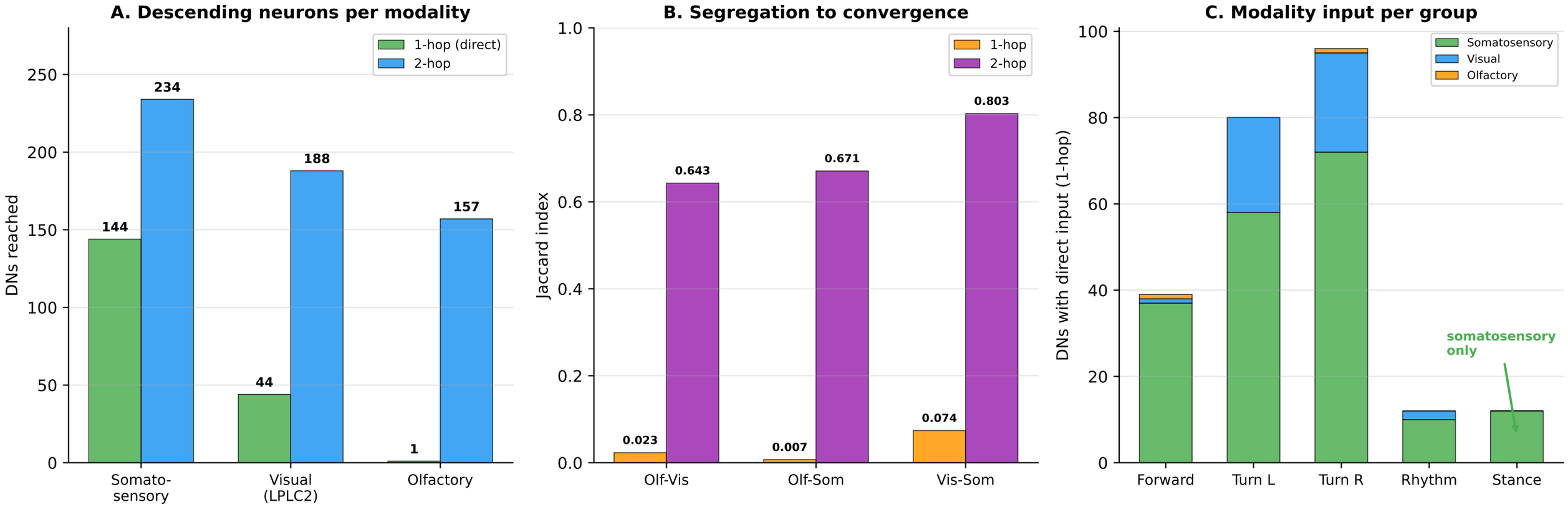

```
[PASS] directional_escape
[PASS] loom_vs_control
[PASS] opposite_turns
[PASS] connectome_specificity
[PASS] statistical_significance

5/5 tests passed
```

Drosophila FlyWire connectome (139k neurons, 15M synapses) | Brian2 LIF simulation | No learning

Figure 5. Modality-specific descending motor channels



Results

2.1 A closed-loop connectome-body interface

We coupled the full FlyWire connectome (138,639 neurons, ~15M synapses) to a MuJoCo biomechanical hexapod via a sensory encoder (body state to Poisson rates for 275-485 identified sensory neurons) and a descending decoder (204-389 DN firing rates to forward/turn/freq/stance locomotion commands). All neurons use identical LIF parameters. No weights were tuned.

2.2 Causal locomotion control (Figure 2)

Ablation of each of 5 neuron groups produced the predicted behavioral deficit:
- Forward neurons silenced: distance -53% (18.9 to 9.0 mm)
- Turn-left silenced: heading shifts rightward
- Turn-right silenced: heading shifts leftward
- Rhythm silenced: step frequency -51%
- Stance silenced: stance gain -33%

10/10 causal tests pass (5 command-level + 5 behavioral).
Shuffled connectome: 3/10 (only trivial contrasts between near-zero signals).
Dose-response: near-linear reduction in forward distance with % neurons ablated.

2.3 Olfactory valence discrimination (Figure 3)

Or42b (DM1, attractive): turn contrast = -0.023 (toward odor)
Or85a (DM5, aversive): turn contrast = +0.044 (away from odor)
Valence contrast = +0.066. 6/6 tests pass.
Shuffled connectome: valence contrast = -0.0005 (eliminated).

2.4 Looming escape (Figure 4)

LPLC2 (210 neurons: 108L, 102R) -> 44 DNs via 1,850 direct synapses.
Loom left: turn = +0.364 (rightward escape)
Loom right: turn = -0.748 (leftward escape)
Escape index = 1.112 (50ms window). Robust across 20/50/100ms.
Shuffled: escape index = 0.053 (21x weaker). 5/5 tests pass.

Results (continued)

2.5 Modality-specific descending motor channels (Figure 5)

1-HOP (direct sensory -> DN):
  Somatosensory reaches 144 DNs (6,990 synapses)
  Visual (LPLC2) reaches 44 DNs (8,823 synapses)
  Olfactory reaches 1 DN (7 synapses)

  Pairwise Jaccard indices:
    Olfactory-Visual: 0.023
    Olfactory-Somatosensory: 0.007
    Visual-Somatosensory: 0.074

  131 DNs exclusively somatosensory. 31 exclusively visual. 0 exclusively olfactory.

2-HOP (sensory -> interneuron -> DN):
  Somatosensory: 5,960 intermediates -> 234 DNs
  Visual: 8,794 intermediates -> 188 DNs
  Olfactory: 1,333 intermediates -> 157 DNs

  Pairwise Jaccard indices:
    Olfactory-Visual: 0.643
    Olfactory-Somatosensory: 0.671
    Visual-Somatosensory: 0.803

  135 DNs reachable from ALL three modalities at 2-hop.

INTERPRETATION: The connectome implements a two-layer architecture:
  Layer 1 (direct): modality-specific labeled lines (Jaccard 0.007-0.074)
  Layer 2 (via interneurons): multimodal convergence (Jaccard 0.64-0.80)

This allows fast modality-specific reflexes (looming escape at 20ms) while
supporting multimodal integration one synapse deeper.

2.6 Functional mapping by modality

  Forward: 37 somatosensory, 1 visual, 1 olfactory
  Turn left: 58 somatosensory, 22 visual, 0 olfactory
  Turn right: 72 somatosensory, 23 visual, 1 olfactory
  Rhythm: 10 somatosensory, 2 visual, 0 olfactory
  Stance: 12 somatosensory, 0 visual, 0 olfactory <-- EXCLUSIVELY somatosensory

Visual input preferentially targets turning groups (escape).
Stance control has zero visual/olfactory input at 1-hop.

Discussion

3. Discussion

We demonstrate that the Drosophila FlyWire connectome, with uniform biophysical parameters and no learning, produces three adaptive sensorimotor behaviors when coupled to a physically realistic body. More importantly, analysis of the descending neurons recruited by each behavior reveals a previously unreported structural principle.

The segregation principle. At the direct sensory-to-descending interface, modalities maintain near-complete segregation (Jaccard 0.007-0.074). Somatosensory, visual, and olfactory inputs reach almost entirely non-overlapping sets of descending neurons. With the addition of a single interneuron layer, this segregation collapses into broad convergence (Jaccard 0.64-0.80), with 135 DNs reachable from all three modalities.

This two-layer architecture has a functional interpretation: Layer 1 supports fast, modality- specific reflexes. The LPLC2 looming escape circuit operates through direct 1-hop connections and produces robust directional escape at 20ms latency. Layer 2 supports multimodal integration for behaviors requiring sensory context.

The stance finding is particularly clean: stance-controlling DNs receive zero visual or olfactory input at 1-hop, exclusively somatosensory. This makes biological sense -- stance gain should be modulated by ground contact feedback, not by distal sensory stimuli. This fell out of the data naturally and was not designed into the decoder.

No previous study has identified this segregation because it requires both a complete connectome to trace all paths AND a behavioral readout to identify functionally relevant DNs. The 175 DNs reached by sensory inputs cannot be distinguished from the 214 unreached DNs by anatomy alone.

Limitations. We use uniform synaptic parameters, a preprogrammed CPG (not VNC connectome), simplified Poisson sensory encoding, and do not model gap junctions or neuromodulation. Despite these simplifications, the system produces robust, stimulus-specific behavior.

Future directions. Extending the segregation analysis to additional modalities, replacing the CPG with a VNC connectome model, and using the dose-response relationship to predict behavioral consequences of genetic silencing experiments.